CHE Student Focus Groups
Spring 2023

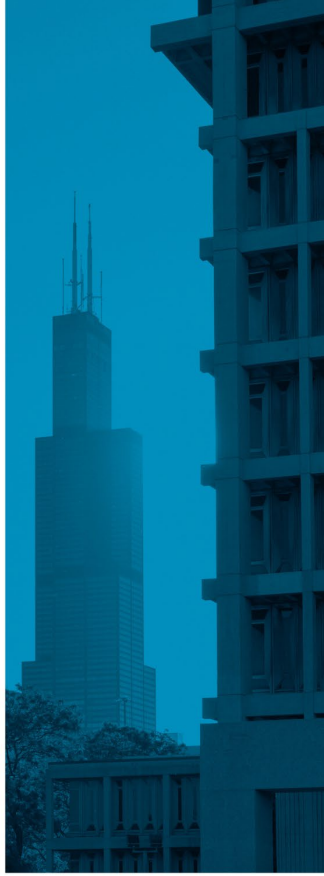


UIC STUDENT AFFAIRS

Focus Groups

- 2 focus groups and 1 individual interview were held in April 2023 (*n* = 17).
  - One focus group had 9 students, the other 7.
  - Each student was compensated $40 for their participation.
  - Students were informed of ethical considerations, e.g., voluntary and confidential.

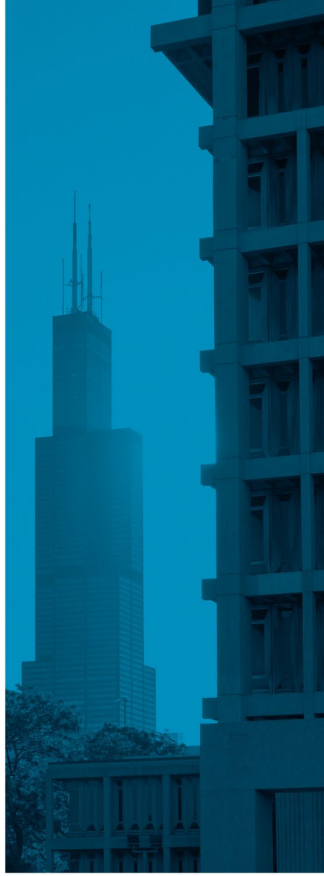
- Focus groups were recorded and transcribed verbatim.
- Recordings will be deleted once fully processed.





Recruitment and Participation

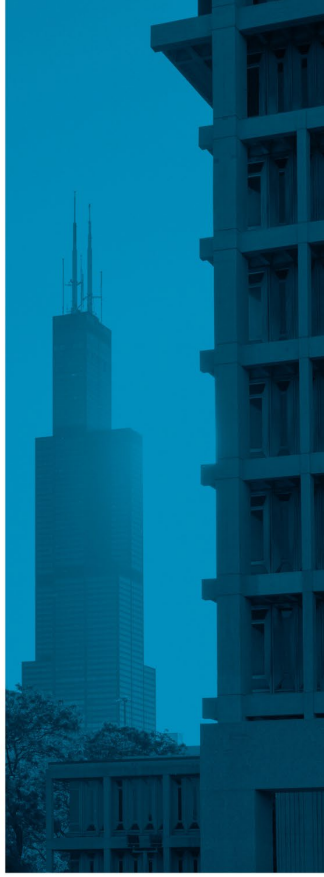
- All CHE students were invited to participate.
- They email Rachel Yim to express their interest and availability.
- Students were assigned to 1 of 2 focus groups based on availability.
- 10 students were assigned to each focus group. Three students did not show or cancelled.





Student Demographic Information

- 65% started at UIC as a first-year student, 35% as a transfer student
- 35% female, 65% male (warehouse does not allow for other options)
- Race/Ethnicity/Status
  - 12% Asian
  - 18% Black
  - 18% International
  - 41% Latinx
  - 12% White
- 24% first generation in college
- 59% started as a first-year CHE student.
- *There were some differences between the two groups as the second group seemed to be further along in the program as a whole.*

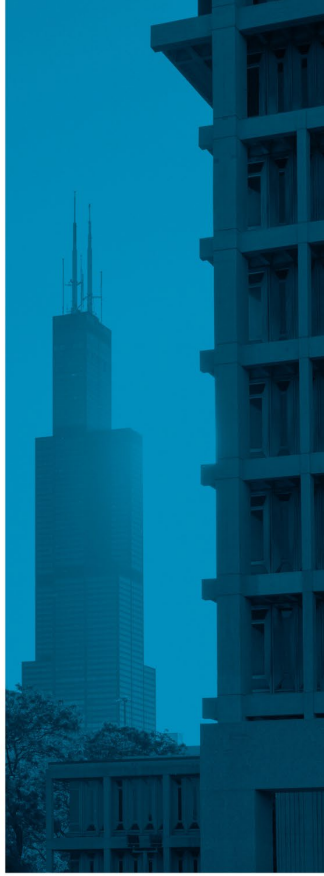




Focus Group Questions

- Students were asked a series of questions to learn about their perspectives on CHE- why they chose the major, courses, climate and support, as well as additional questions on spaces and resources.
- Data presented are key themes that emerged from the focus groups.
- There are more data than what I will be presenting; focusing on feedback for things to consider for work of the DAT.
- Some of feedback is applicable to individual faculty and some is more directed toward the department.

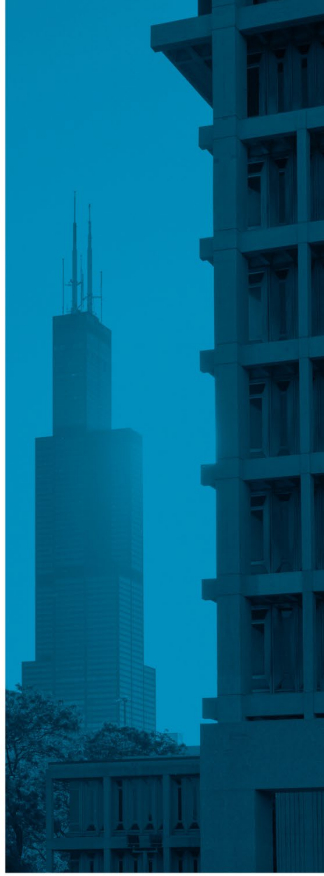
- *Students in the two focus groups reported very different experiences in some areas.*





The Ice Breaker: Pathways into CHE

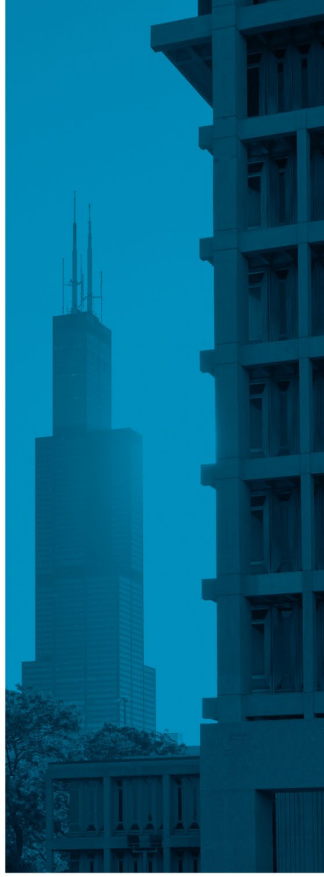
- Students liked the blending of chemistry and “hands on” (engineering)
- Parents liked engineering
- Good paying jobs
- Career flexibility: Would be able to work in a range of fields





Course Content and Sequence

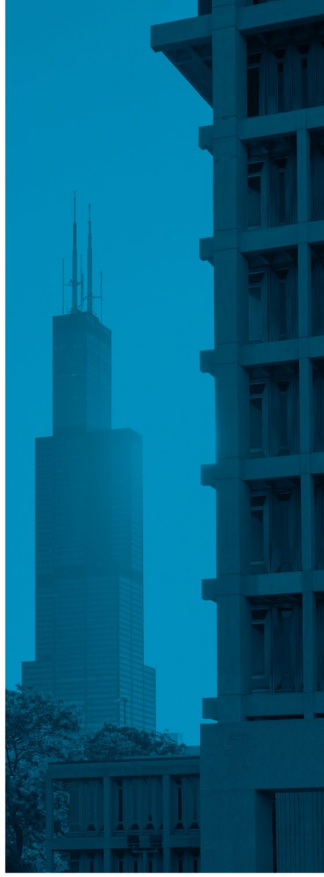
- Course workload is too much, particularly 3rd year.
  - Amount of work per class X multiple classes in major makes it very challenging to keep up.
  - Homework assignments, in particular, can be too much.
  - Course credits don't align with workload expectations.
  - *Consider: mapping out expected time for classes and homework for a typical courseload.*
- Earlier years are heavily theory, less hands on. Would appreciate more hands on in earlier classes.
  - *Consider: Work with CATE to develop active learning strategies for classroom lecture.*





Course Content and Sequence continued

- Course sequence works well for most in terms of content.
  - Some small disconnects, but generally positive.
- Courses need to be taught both semesters.
  - If you don't take a class or fail a class, will put you behind a full year, not just the semester.
  - This is particularly challenging for transfer students as they have less time to manage if they are behind.
  - Financial aid/scholarships may run out if not complete on time.
- Many of the required non-CHE course are not needed. Students suggested looking at course requirements to see if they are all needed for the degree.

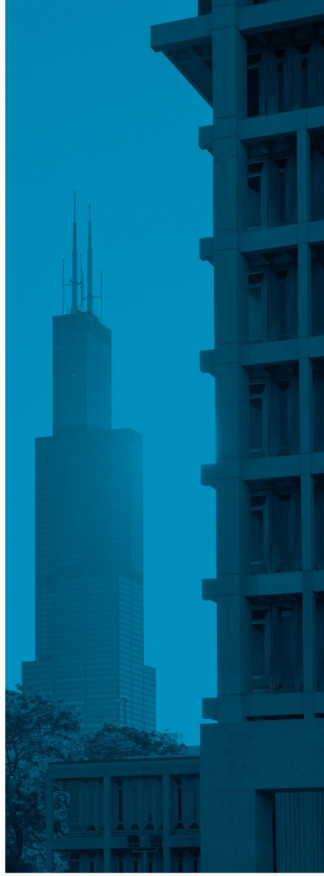




Academic Resources Needed

Students made specific suggestions to support their academic success:

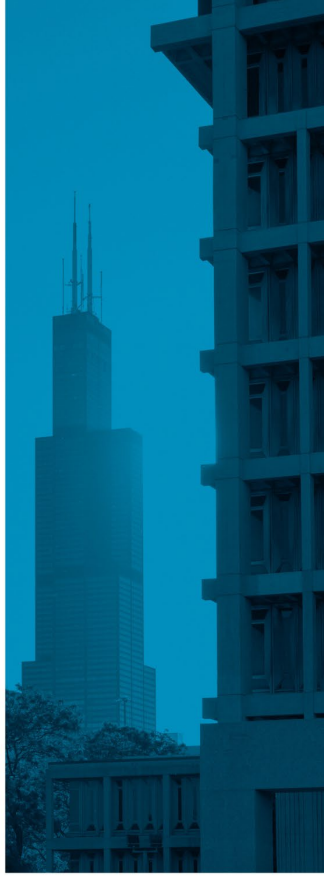
- Faculty office hours are not enough, would like time available each week. Consider offering at different times of the day.
- Need more TA hours too.
- Consider putting lecture notes on Blackboard.
- CHE-specific tutoring is needed.





Sense of Belonging

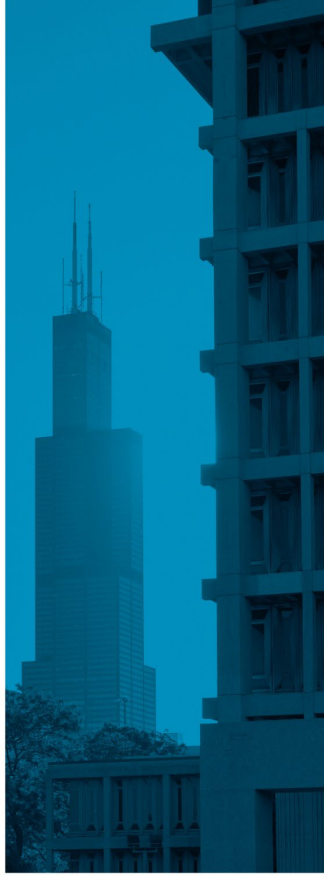
- Most of the students to some degree feel like they didn't belong, at least at times.
- Feel like they must constantly prove to others, particularly faculty, that they are meant to be there.
- They often continually convince themselves that they belong.





Sense of Belonging

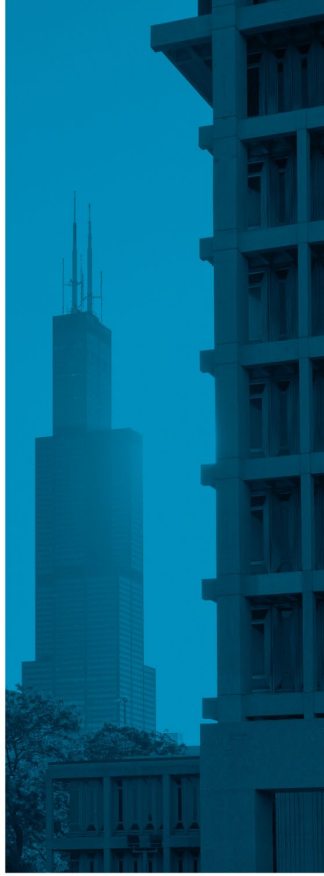
- Get positive feedback from students outside the college about being capable- told they are smart because in CHE.
- Some professors are intimidating until they get to know them.
  - Fear of the unknown; hear what other students say.
- *Previous research insight: high school messaging is that faculty don't care.*
- Not doing well in classes makes them question themselves and if they should be there.





Faculty behaviors that promote and discourage students feeling like they belong

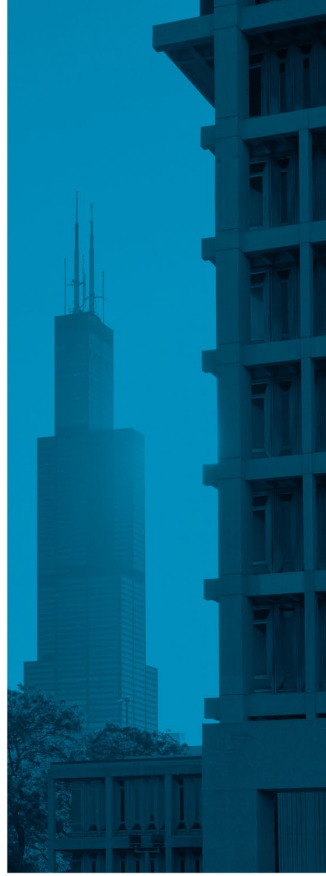
- Faculty create a welcoming environment in the class, particularly the first day.
- Given engineering is male dominate, having women faculty helps female students.
- Faculty are not always “present”; leave right after class.
  - *Consider building in time after class to be available.*

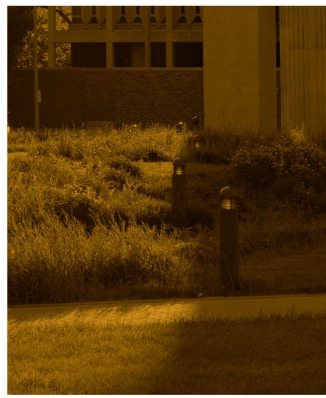




Faculty Behaviors that Promote and Discourage Students Feeling Like They Belong

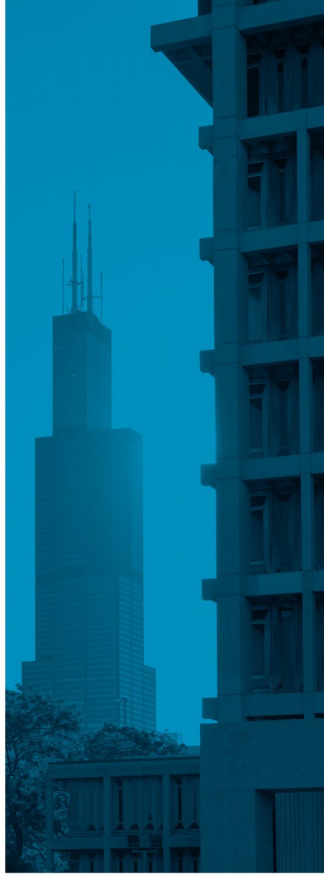
- Some students don't feel emotionally safe in some classes.
  - Feel judged for not knowing content.
  - Make the experience of failing even worse, not a motivational or learning experience.
    - => makes it harder to get help as students don't want to ask question or go to office hours.
  - *Consider: CATE has a lot of resources to help develop skills and strategies to promote belonging in the classroom.*





Faculty Behaviors that Promote and Discourage Students Feeling Like They Belong

- Students not allowed to/or have to ask to use bathroom or get water; feel shamed if they go.
- No exception attendance policy.
  - Life challenges come up; finances can get in the way.
  - *Previous research: Sometimes have to make decisions about finances versus school; some students have family obligations that disrupt ability to attend class at times.*
- Asking for “proof” to excuse late work in front of other students is embarrassing.
- *Recurring theme about privacy for personal issues and feeling of embarrassment.*

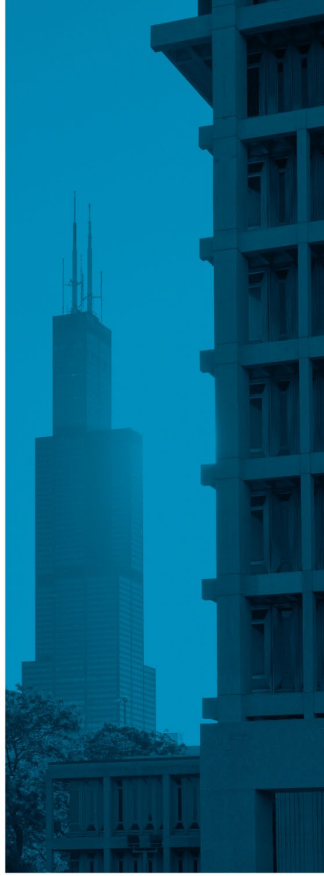




Teaching Assistants/Graduate Students

- TAs are generally very helpful, very positive feedback.
  - Often see them as role models.

- On occasion, they are not familiar enough with course content.

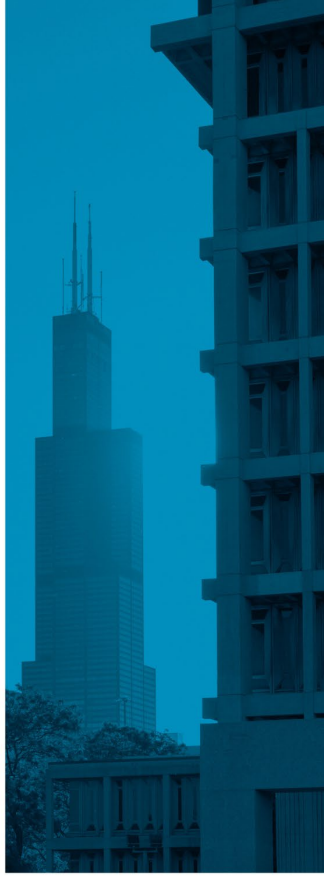




Sense of Community Among Students is Complex

- *The two focus groups had very different feedback in this area. The difference may be due to year in school and entry to program (first-year student vs. transfer student/inter-college transfer) or maybe one group was less supportive to other students.*

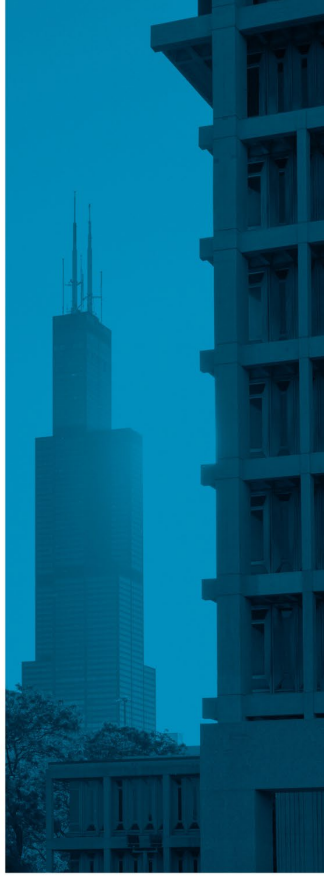
- FG1: Students have strong community and support each other.
  - Program is hard, but they struggle together and support each other.
  - More advanced students are helpful to new students.
- (FG2): Many peers are “toxic”, particularly the more advanced students.
- Transfer students and inter-college transfer students have a hard time connecting with other students.





Sense of Community Among Students continued

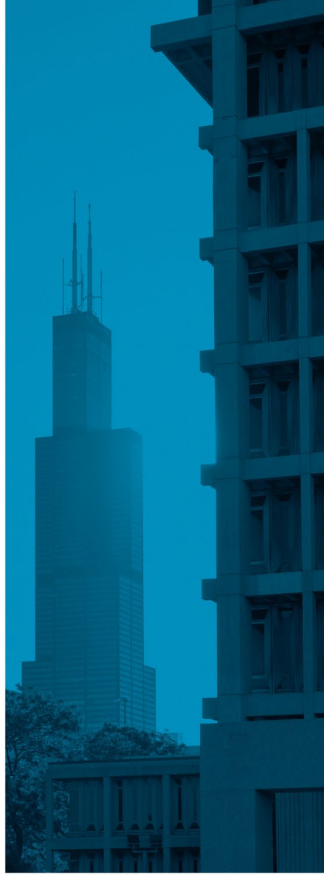
- First-year should provide more opportunities for students to connect.
  - Some students feel like they are excluded if they aren't part of "the group" that connects in ENG 100.
- Hard to connect with students outside of the department due to workload (for many, not all).
  - Had more involvement with campus and other students early on but had to stop due to workload.
- *The first-year comes up often as a place that is potentially challenging for students. May want to consider ways to increase engagement and belonging in first year and also repeat those opportunities for transfer students.*





Advising is Inconsistent

- Some students report that advising is excellent, but this is not consistent feedback. *Time seems to be a driving factor on feedback.*
- Some specific challenges:
  - Many reported multiple advisor changes, this creates difficulty and advising experience is not consistent.
  - Some not told to take correct classes and then are behind.
  - Need more advising in the first year, particularly around program structure.

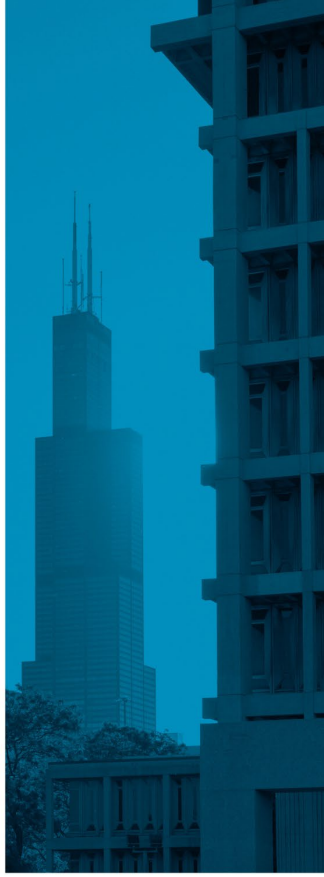




Advising continued

- Advising sessions
  - Not enough time spent: shows form and told to pick; students interpret this as advisors don’t care and want session over.
  - They do not like zoom advising, especially with the camera off.
- Include ongoing conversations about career paths, discuss their strengths, and ask what they like.

- *Consider: Faculty PD around advising, what to include, strategies for engaging students and making them feel welcome, etc. Can create resources for faculty (e.g., checklist) on what to cover.*



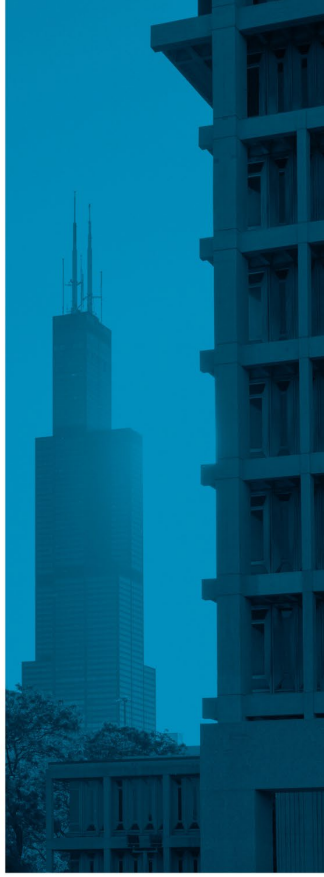


Health and Well-Being

- Some students don't have enough to eat
  - Finances make this difficult
  - No time as options are far away
  - Stress leads to loss of appetite

- Stress of coursework and feelings of failure impacts mental health.

- *Consider: Sharing resources for basic needs (Dean of Students) in syllabus or other ways to normalize this and not point out specific students.*

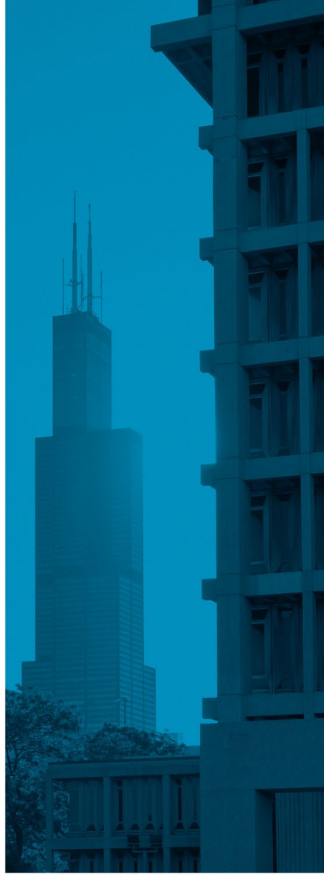




The Building

- Being in the building creates belonging.
  - Like seeing faculty in the building, talking to them informally.
  - Helps building community among the students.
  - *Consider: create ways for students to be in the space their first-year.*

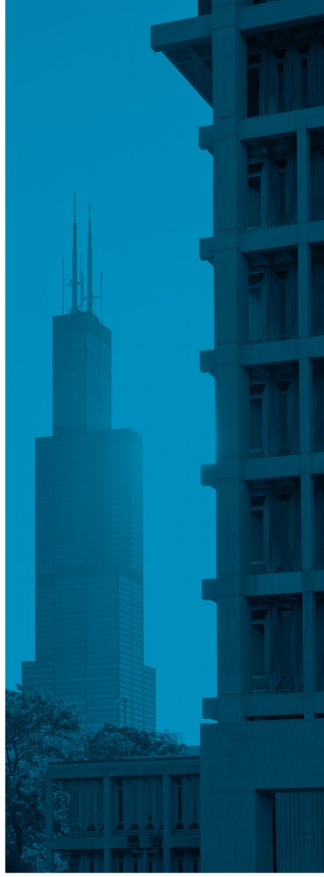
- Student suggestions for improvements:
  - More places to study in the building.
  - (Fix) vending machines with more/better food options.





High Impact Practices and Engagement Experiences Promote Belonging and Skills Development

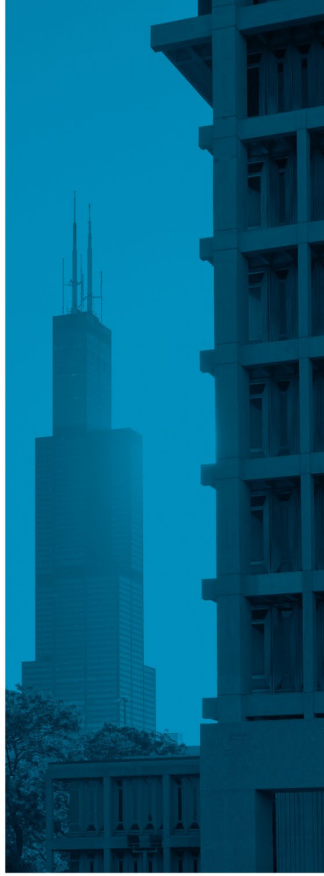
- Working in Research Labs helps students feel like they belong in CHE
  - Helps them connect to faculty
  - Exposed students to graduate students who can also serve as mentors
  - More students were interested in this than were involved
  - *Consider: Increasing opportunities to engage in research, be pro-active by inviting students in, consider using CURA for work study students (may also help with finances).*
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High Impact Practices and Engagement Experiences Promote Belonging and Skills Development continued

- Internships helps students feel like they belong in the field.
  - Internships with **alumni** would be particularly beneficial (for international students).
- AIChE- brought up by multiple students with very positive feedback.
  - Allows students to get to know faculty.
  - Like the events.
  - About half of the students had engaged.
  - *Consider ways to increase student involvement.*

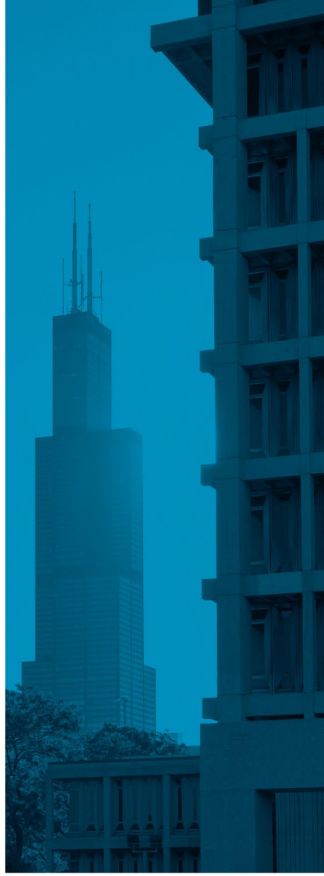




Diversity, Equity and Inclusion

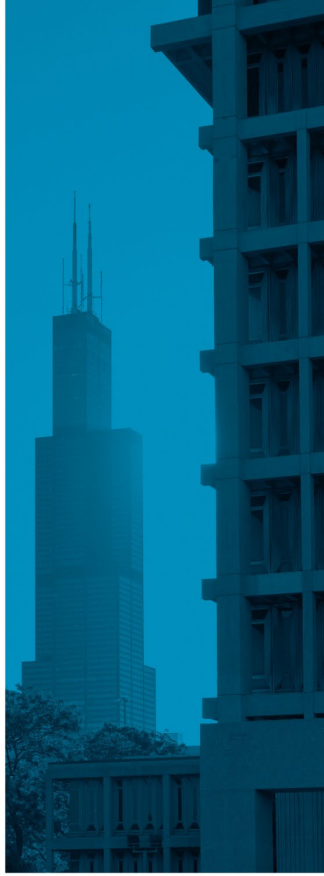
- Mixed feedback from students, some very positive, others with constructive feedback.
- Need more queer faculty and TAs.
- Black students are underrepresented. It is difficult to be in a class where they are very underrepresented.
- Some students report toxic behavior by peers related to religion that is not always addressed by faculty.

- Some students felt that efforts were responsibility of students (DEI groups); want faculty to prioritize and take lead.





Questions? Comments?





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